Reliable Collective Reasoning and Evaluation for AI Agents

Overview AI systems increasingly use teams of language models to interpret evidence, critique solutions, and answer difficult questions. Combining several assessments can be more reliable than trusting one model, but agreement alone is weak evidence of correctness. Multiple agents may reproduce an error from the same source, share training-induced biases, or accept a persuasive but invalid reasoning step. In these cases, majority voting or unconstrained deliberation can increase confidence without adding information. This problem becomes more acute when agents have heterogeneous expertise, evidence is incomplete, and independent ground-truth labels are unavailable.

We propose a framework for evidence-grounded collective reasoning in which agents expose checkable intermediate claims, their supporting sources, and the uncertainty of each inference. Three methodological challenges motivate this research: (1) evaluating the validity of diverse reasoning processes without relying solely on final answers; (2) inferring reliability when expert judgments and cited evidence are statistically dependent; and (3) deciding what new evidence or targeted check will most efficiently resolve collective uncertainty. The research integrates fine-grained reasoning evaluation, dependence-aware aggregation, and active verification rather than treating model votes as interchangeable observations.

We will pursue three interrelated research threads:

- **Process-grounded individual assessments:** We will decompose solutions into intermediate assertions and link them to retrieved documents, structured relations, and logical premises. Synthetic multihop reasoning examples and controlled factual, contextual, and logical errors will support the training of fine-grained evaluators. We will study which correctness judgments can be calibrated on reference cases, how expertise transfers across task types, and how to distinguish evidence-backed reasoning from merely plausible explanations.
- **Reliability and dependence in collective inference:** We will estimate agent expertise and the incremental information conveyed by a new judgment while accounting for shared source provenance, common reasoning patterns, and correlated errors. Building on our work in Peer Rank and Discussion, we will investigate uncertainty-aware fusion of peer assessments, including cases in which agents agree for the wrong reason. A central question is when correlations are identifiable from observations and when the system must abstain or request external evidence.
- **Active verification and adaptive deliberation:** Instead of inviting indefinite discussion, the system will choose targeted actions: retrieve an independent report, challenge a disputed inference, test an assumption with a tool, or solicit an agent with complementary expertise. We will learn policies that balance expected reduction in uncertainty with verification cost and revise the collective belief only when new information warrants it.

Keywords: Collective reasoning, Multi-agent evaluation, Uncertainty, Dependence-aware aggregation.

Intellectual Merit The project will develop new algorithms for process-aware reliability estimation, principled treatment of correlated AI judgments, and adaptive evidence acquisition. We will evaluate accuracy, calibration, failure detection, abstention, and verification efficiency against single judges, majority voting, learned weighting, and discussion baselines under controlled common-source errors.

Naval Relevance and Broader Impact The resulting methods could strengthen multi-source analysis and human-supervised decision support relevant to Naval information environments, in which apparent consensus may conceal duplicated reporting. We will release benchmark tasks and evaluation tools based on public and synthetic evidence, without assuming access to operational Navy data.